



Linux+

Linux



- Linux is Open Source Operating system that can be modified by you
- Anyone can add , remove code in linux

Linux Distributions

There are many distribution for linux :

Debian (Kali)

Ubuntu

Red Hat

Fedora

Centos ...

Linux

- Linux is used for android mobiles as Operating system also
- More secure than Windows
- More Fast than Windows
- Provides all Hacking tools For Free

Linux



- Installation:
- There are Three types of installation :
- 1- Primary System
- 2- Virtual System
- 3- Bootable USB

Linux

- For Primary System you need to install it without any other Operating System
- (like Windows , Mac Os)

In Primary System , Device would get into Linux directly after boot

Linux



- Just go to kali.org then go to installer images then download 64 bit Image
- Then Burn Iso Image on USB by [PowerIso App](#) or [balenaetcher](#)
- Then shutdown your device and put USB and f12 or f10 or ESC to get the boot
- Choose USB and start installing

Linux



- For Virtual Installation you need VMware
- Then install Iso image or VDisk Image
- And continue installation

Linux



- For bootable USB you don't need to install linux on device , just install it on USB and then put it on your Laptop and it will open the linux system directly
- All what you need go to kali.org and choose bootable USB image
- You can install another image called tails
- Then burn it on USB by [balenaetcher](#)

Linux



- After installing the system open terminal and update it by :
 - `# sudo apt update`
 - `# sudo apt upgrade`
 - apt is used to install new applications in linux
 - Sudo means super user or root (high privileges)
- (When you see (Permission denied) use `# sudo`)

Linux Commands

- All commands are small letters
- To be root
- # sudo su
- And to be normal user
- # exit
- To see the current user
- # whoami

Linux Commands

- To see files on same directory or folder
- # ls
- To see the current folder
- # pwd
- To move to folder xyz
- # cd xyz

Linux Commands

- To get back to previous folder
- # cd ..
- To read file called xyz.txt
- # cat xyz.txt
- To create new file called xyz.txt
- # nano xyz.txt then write then CTRL+X then Y

title

- To remove file called xyz.txt
- # rm xyz.txt
- To create new folder called xyz
- # mkdir xyz
- To remove the folder xyz
- # rm -r xyz

Linux Paths

- There are two types of paths :
- absolute path starts with /
- Ex: /home/hossam/downloads/
- Relative path :
- Is it current path and you don't need to describe the full path

Linux Path

- If you are in folder `/home/hossam/downloads` and need to remove file called `xyz.txt` in folder `downloads` then you don't need to write the full path
- `# rm xyz.txt`
- But if you are in different folder than what you need to access then you will write the full path
- `# rm /home/hossam/downloads/xyz.txt`

Linux Commands

- To move file xyz.txt from place to /home directory
- # mv xyz.txt /home
- To change the name of xyz.txt to rednexus.txt
- # mv xyz.txt rednexus.txt
- To copy file from place to /home directory
- # cp xyz.txt /home

Linux Commands

- Difference between write and append
- To write is to add new content to file but deletes the old content >
- To append is to add new content without deleting old content >>

Linux Commands

- To print word rednexus in terminal
- # echo "rednexus"
- To write word "rednexus" inside file called xyz.txt
- # echo "rednexus" > xyz.txt
- To append word "rednexus" inside file called xyz.txt
- # echo "rednexus" >> xyz.txt

Linux Commands

- To get information about network
- # ifconfig
- To get information about memory space
- # free
- To get information about hard disk
- # df -H
- To get information about processor
- # ps aux

Linux Commands

- There are many ways to install applications in linux
- First : use `# apt install <application>`
- Second: use `# snap install <application>`
- Third : use `# dpkg -i <application.deb>`

Linux Permissions

- Permissions for files are (read , write , execute)
- Execute $\Rightarrow 1 \Rightarrow x$
- Write $\Rightarrow 2 \Rightarrow w$
- Read $\Rightarrow 4 \Rightarrow r$
- If you need read & write then $4+2 = 6$
- If you need write& execute then $2+1 = 3$
- If you need all read& write & execute then 7

Linux Permissions

- To see the permissions of files and folders
- # `ls -lah`

Linux Permissions

```
-rwxrwxrwx 1 hossam hossam 12K Apr 21 17:16 swisscom_notes.txt
```

- First place - means that this is not a directory
- Remains 9 places let's separate every 3 places
- The first 3 rwx means read & write & execute but for the **owner**
- The second 3 rwx for **group users**
- The third 3 rwx for other people who can access

Linux Permissions

```
drwxr-xr-x  3 hossam hossam 4.0K Jun 22 11:53 .ghauri
drwxr-xr-x  3 hossam hossam 4.0K Aug 15 2022 .gnome
```

If you realized the first bit found (d) which means that is directory

And the First three bits are (r w x) => for the owner

The second three bits are (r - x) => for group users

The third three bits are (r - x) => for any other user

Linux Permissions

- To change the permissions for a file xyz.txt
- # `chmod 777 xyz.txt`

- To add execute permission to a file xyz.txt
- # `chmod +x xyz.txt`

Linux Piping and redirect

- To redirect the output to another command use |
- To get the word hossam from file called xyz.txt
- # cat xyz.txt | grep "hossam"
- To know the number of lines in file xyz.txt
- # cat xyz.txt | wc

Linux Labs

- You can get vps
- # ssh root@segfault.net
- And the password: segfault
- <https://overthewire.org/wargames/bandit/bandit0.html>
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- [Hackthebox](#)

Linux



RED
nexus

The end